TÉCNICO LISBOA

Computational Vision

Project 1 Guide and Statements

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Contents

Project 1 Evaluation Guide **2**

Report Structure and Presentation 3

Report Contents, Implementation and Analysis 3

Provided Code and Interpretability 4

Project 1 Statements **6**

Topic A: Wildfire detection from aerial images 8

Topic B: Railway inspection using on-train cameras 9

Topic C: Gate segmentation for racing drones 10

Topic D: Cone detection for FST competitions 11

Topic E: Wind turbine inspection using drones 12

Topic F: Earth surface characterization from aerial images 13

Topic G: Skin lesion segmentation 14

Topic H: Historical architectural surface crack detection 15

Project 1 Evaluation Guide

Four different aspects contribute are part of the assignment criteria, each one with the following relative weight in the overall grade:

- Report Structure and Presentation: 4/20
- Report Contents, Implementation and Analysis: 10/20
- Provided Code and Interpretability: 3/20
- Self and peer review: 3/20

In turn, these components are are detailed into two or more parts, accounting for the multiple aspects to be evaluated in the report, as illustrated by Table 1, shown below. The description of each of these sub-components is provided next. It is strongly recommended to carefully review and understand the assessment criteria, not only for for the development of your report, but also to correctly evaluate the ones from other groups.

Table 1: Assessment criteria for Project 1

Report Structure and Presentation		Report Contents, Implementation and Analysis				Provided Code and Interpretability			Self and peer review		Overall
General aspects	Report structure	Approach	Methods and tools	Tests, results and analysis	Conclusions	General use	User-friendliness code commentaries	Modularity	Self review	Peer review	
2	2	3	2	4	1	1	1	1	1	2	20

It should be noted that the project final grade will take into account the self and peer review assessments, as well as the evaluation by the faculty.

1. Report Structure and Presentation

Overall weight in grade: 4/20.

- General aspects (2/20):
  - The report should be written in academic English, and obey standard orthographic and grammatical rules.
  - The images generated should be placed at the right of the code in the live script, to avoid unnecessary extension of the document, and to facilitate the follow-up of the code. The images used for contextualization and/or to explain formulas can be placed below the text or at the right.
  - Every image should ne numbered and have a title, using the numbering to mention them throughout report.
  - Together with the live script, the PDF version generated by it is mandatory, to ensure that the evaluation can be made even if an issue with the code occurs.
- Report structure (2/20):
  - The report should be structured using a standard structure (*IMRAD* - Introduction, Methods, Results, Analysis and Developments). More sub-sections or an adequate variation can be used if needed, and conclusions should also be included.

2. Report Contents, Implementation and Analysis

Overall weight in grade: 10/20.

- Approach (3/20):
  - In the report, the strategy for tackling the problem should be well-explained and justified. Here, a delineation and justification of which image processing tools were used are expected, also explaining how they are chained in order to form a pipeline of development (pre-processing, main and postprocessing techniques), from the starting image (or images) to the final result. If there are multiple approaches that could work, describe them accordingly, explain why each was chosen or not, and what visual characteristics they exploit. Note that, if it is suitable to the problem at hand, multiple approaches could be envisioned to succeed in solving it, and eventually a comparison may be drawn.
- Employed methods and tools (2/20):
  - After delineating the algorithms in the approach, the next step is to state and describe the methods of each step and how you tuned the respective parameters. For example, if one of the steps requires thresholding, explain, in generic terms, what method was applied, why and how.

- Tests, results and analysis (4/20):
  - Once the approach has been described and the underlying methods to be used have been explained, the results of the implementation should follow. If a certain method was used - for example, selecting particular images and/or grouping them in development and test groups - this should be clear. **Performance metrics** should be provided, define the appropriate ones for your project. The results should be provided and, assuming these are shown in an image format, they must be adequately mentioned the text. Lastly, an analysis about the results should be made, and is expected to be comprehensive, exploring the benefits and limitations of the undertaken approach and methods.
- Conclusions (1/20):
  - Following the analysis regarding the performance of the developed solutions, the conclusions should summarize such results, reflecting on their success, limitations, and pointing to possible directions of improvement as future work.

3. Provided Code and Interpretability

Overall weight in grade: 3/20.

- General use (1/20):
  - A code implementation is suggested since virtually all the tools used for image processing in this course are directly available, but Simulink can be used instead, if desired. Nevertheless, if this option is chosen, a live script should still be provided as a report, executing one or more Simulink models which must redirect their output to the live script.
- User-friendliness, code commentaries (1/20):
  - The provided code should be user friendly: the names of the variables should be chosen according to their meaning and use in the code; the code should be commented, explaining the purpose of each function (MATLAB or user-defined) used; and it should be organized by sections, preferably according to the designed algorithms. If a Simulink implementation is used instead, the blocks should be named similarly to their function in the development pipeline, and usage of text boxes for added information is recommended, aiming to have the same purpose as commentaries in a code implementation.
  - Every line of code that is commented is assumed to be a commentary, thus not contributing for the overall code. Therefore, legacy or residual code should be avoided.
- Modularity (1/20):
  - The usage of user-defined functions is strongly recommended, specially when repeating certain methods or tools. For example, if a filtering operation followed by binarization is used

several times along the code or for a group of images, these two operations can be defined as a function that takes an input image and a set of parameters (filter intensity, for instance) and returns the filtered and binary image. Functions should be defined in external files and not in the live script, and must be called directly by this script. Naturally, functions should also have code commentaries describing how they work, which inputs they take and the outputs they return.

4. Self and peer review

Overall weight in grade: 3/20.

- Self review (1/20):
  - After submission of the project, you will be asked to grade your own project (limited to 17 out of 20), according to the aforementioned assessment criteria. The capacity to analyse your own work and evaluate it will be checked, and also be a part of the final grade as a self assessment component. Provide an adequate justification of your quantitative assessment with a qualitative analysis regarding positive and to improve aspects of your work. For each criteria you do not give the maximum grade, please provide a comment justifying what is missing in the respective cell.
- Peer review (2/20):
  - The previous task should be repeated for evaluating two reports from your peers, which is done anonymously. Once again, the ability to analyse and grade the projects will be checked, helping you gain an understanding about CV applications other than the one of your project. Please provide the assessed groups a qualitative assessment based on constructive feedback regarding positive and to improve aspects of the work, in agreement with the quantitative assessment you made. For each criteria you do not give the maximum grade, please provide a comment justifying what is missing in the respective cell.
- A detailed Excel sheet for the self and peer review will be provided to each group. The self and peer review grade is based on the quality and completeness of the assessments.

Project 1 Statements

The main objective of this project is to exploit different solutions to achieve the proposed goal. It is not necessary to find a definitive solution, but only to gain insight in how easy/difficult the problem can be. Therefore, the working time dedicated to this work should pay attention to this point and as expected, the final mark of the work will have in consideration more the effort put to exploit approaches to the problem, including feasibility and weak points, than to the quality of the results. It should be noted that this work covers a limited part of the topics of this course, and naturally does not replace the work proposed to the labs. Pay attention that part of the exam questions may be based on the work proposed in the both the projects and the labs.

Report format and software version

The report must be submitted in a ***Live Script***, and it is suggested to become familiar with this tool. Recommended sources for an introduction to *Live Scripts* can be found in the following links: Link 1, Link 2, Link 3. If Simulink is chosen for the implementation, it must be called from the *Live Script* and the resulting image/s routed back to it, so they can be displayed.

All the proposed problems are to be solved using **2023a** and the **correspondent Simulink version**. Other versions of MATLAB may not include some functions, or these may vary in the input/output arguments, and therefore all the delivered files, namely the live script, Simulink models and/or additional functions should run using this version.

Submission instructions

The project must be submitted in Fenix as a zip file, having inside the MATLAB 2023a live script file and any other MATLAB 2023a or required files to run it. If, for any reason, you wish to join a film or a large set of images, those files may be emailed via link with unlimited download time. The name of the project zip file to be submitted in Fenix should follow the pattern:

G<group#>_<student#1>_<student#2>_<student#3>.zip
e.g., “G10_11111_22222_33333.zip”.

Submission deadline: 18/3/2026

Self and peer review guidelines

Once the submission deadline has passed, you will receive an Excel table with fields corresponding to the multiple assessment criteria. You will then evaluate your own report accordingly, as well as the ones from two other groups. As stated before, the ability to analyse and evaluate the reports from other groups, as well as your own, is also part of your own evaluation.

Self and peer review deadline: 26/3/2026

Topic A: Wildfire detection from aerial images

One of the issues of wildfires firefighting in hilly landscapes is to know with, an acceptable accuracy, where the fire is, so as to know where the operational teams should go. Aerial images are of great help and some automatic detection and localization would be highly welcome. In the images of typical interior landscapes, this work should be able to answer on whether a fire is active or not and locate it in the image. This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 1.

Figure 1: Sample image from the dataset of Topic A.

Problems A1 and A2

For the given datasets A1 and A2, it is intended to determine an automated procedure for wildfire detection:

1. Observe the images and organize them in folders according to the presence or not of fire (do not change the filenames).
2. Design and implement an algorithm to segment fire, when existing.
3. Analyze the possibility of automatically classifying the image regarding the existence of fire.
4. Exploit image enhancement techniques with the objective to make more evident the fire location.
5. Compare your automated fire detection procedure performance against expert annotations (yours, for selected images) using standard metrics (justify your choice of performance metrics).
6. Analyse failure cases and discuss which image characteristics most challenge your methods. Beware of possible false negatives and false positives. Which do you consider more important to minimize?

Topic B: Railway inspection using on-train cameras

Trains have the great advantage of a segregated path so that they may move at high speed with no safety issues. However, this is only true if no obstruction appears on the rails and it is quite important to detect whether the rails are indeed free or if some object like a fallen tree or rock is blocking the passage. An automatic inspection of the rails in front of the train is proposed here.
This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 2.

Figure 2: Sample image from the dataset of Topic B.

Problems B1 and B2

For the given datasets B1 and B2, it is intended to determine an automated procedure for railway obstruction detection:

1. Compare the images and characterize the rails within each of the images.
2. Design and implement an algorithm to segment the image and isolate the rails. How can you make use of the Hough transform?
3. How can you confirm the rail ahead is free?
4. Exploit image enhancement techniques with the objective to make more evident the position of the rail and obstacle.
5. Compare your automated railway obstruction detection procedure performance against expert annotations (yours, for selected images) using standard metrics (justify your choice of performance metrics).
6. Analyse failure cases and discuss which image characteristics most challenge your methods.

Topic C: Gate segmentation for racing drones

Drones have been increasingly popular as a research application topic as well as for games or competitions. For such a competition, multi-rotors are supposed to go through a sequence of gates and vision is the natural way to localize these gates and then guide the drone through their center. This dataset includes a few images taken during such a competition and the work is to automatically try and find the gate within each image.

This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 3.

Figure 3: Sample image from the dataset of Topic C.

Problems C1 and C2

For the given datasets C1 and C2, it is intended to determine an automated procedure for gate identification:

1. Design and implement an algorithm to segment the image, isolating the gate.
2. Explore different image processing techniques to localize the gate extreme points and centroid. How can you make use of the Hough transform?
3. Exploit image enhancement techniques with the objective of making more evident the position of the gate.
4. Compare your automated gate detection procedure performance against expert annotations (yours, for selected images) using standard metrics (justify your choice of performance metrics).
5. Analyse failure cases and discuss which image characteristics most challenge your methods.

Topic D: Cone detection for FST competitions

The formula student is an example of integrated work and competition where students are asked to show their capacity in designing and assembling a prototype with interesting and advanced features. Next generation of cars are expected to be capable of autonomous driving and vision is certainly an adequate tool to help in the perception of the obstacles. This work is exploring a simplified scenario where the road is identified by cones on each side and these cones should be automatically detected and tracked in order to steer the car to the middle of the road.

This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 4.

Figure 4: Sample image from the dataset of Topic D.

Problems D1 and D2

For the given datasets D1 and D2, it is intended to determine an automated procedure to detect cones:

1. From the videos in the dataset, extract typical frames as representative images to work with.
2. Design and implement an algorithm to segment the image and isolate the cones.
3. Analyze the possibility of automatically detecting cones in an image and tracking them. Notice you have cones with different colours, they should also be classified by colour.
4. Exploit image enhancement techniques with the objective to make more evident the position of the cones.
5. Compare your automated cone detection procedure performance against expert annotations (yours, for selected images) using standard metrics (justify your choice of performance metrics).
6. Analyse failure cases and discuss which image characteristics most challenge your methods. Beware of possible false negatives (not detecting a cone) and false positives (falsely detecting a cone). Which do you consider more important to minimize?

Topic E: Wind turbine inspection using drones

Commercial-grade UAVs handled by professional operators can provide visual inspections with a higher resolution than using ground-based inspections. They also provide an inspection that is quicker, easier, and less costly and risky than rope access techniques. Current practical applications can be found, for example, in:

- UAV Drone Wind Turbine Inspection - YouTube
- Drones for blade inspection of wind generators - YouTube

This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 5

Figure 5: Sample image from the dataset of Topic E.

Problems E1 and E2

For the given datasets E1 and E2, it is intended to determine an automated defect procedure:

1. Search the internet to add some more information about this inspection process besides that already given (e.g. see reference).
2. Signalize and characterize the defects that are present in the provided images.
3. Design and implement an algorithm to segment the blade from the remaining of the image.
4. Exploit different image processing techniques for the defect detection.
5. Analyse the possibility of automatically classifying the different types of defects.
6. Exploit image enhancement techniques with the objective to make more evident the presence of defects to a human operator.
7. Compare your automated defect procedure performance against expert annotations (yours, for selected images) using standard metrics (justify your choice of performance metrics).
8. Analyse failure cases and discuss which image characteristics most challenge your methods.

Topic F: Earth surface characterization from aerial images

Currently, earth surface characterization from aerial images obtained using drones has a huge set of applications. For fire prevention and fighting it is important to recognize the type of existing coverage, like, grass, water, roads or paths, buildings, etc. This work intends to constitute a contribution based on a set of provided images, exploiting studied techniques.

This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 6.

Figure 6: Sample image from the dataset of Topic F.

Problems F1 and F2

For the given datasets F1 and F2, it is intended to:

1. Analyse the images provided in the dataset and characterize them regarding the different parts to detect or segment (trees, roads, buildings).
2. Segment each image in superpixels and study the influence of the tuning parameters in the result.
3. Design a strategy for merging superpixels if they correspond to the same type of coverture.
4. Analyse the possibility of automatically segmenting different types of coverture, eventually proposing a set of descriptors to be used in this process.
5. Compare your automated terrain classifier against expert annotations (yours, for selected images) using standard metrics (justify your choice of performance metrics).
6. Analyse failure cases and discuss which image characteristics most challenge your methods.

Topic G: Skin lesion segmentation

Early detection of melanoma significantly improves patient survival rates, but accurate diagnosis requires precise delineation of lesion boundaries in dermoscopic images. This project uses the PH^2 database, developed by the University of Porto's ADDI (Automated Dermoscopic Diagnosis of Images) group (check reference). The dataset contains 200 dermoscopic images acquired at the Dermatology Service of Hospital Pedro Hispano in Matosinhos, Portugal. The database includes 80 common nevi (benign moles), 80 atypical nevi (suspicious lesions), and 40 melanomas (malignant). All images are 8-bit RGB (768 × 560 pixels) from Fitzpatrick skin types II-III (white to cream white), with expert-annotated manual segmentation masks provided by dermatologists. Each lesion is also clinically classified and annotated with dermoscopic structures, including pigment networks, dots, globules, and streaks.

This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset is shown in Figure 7.

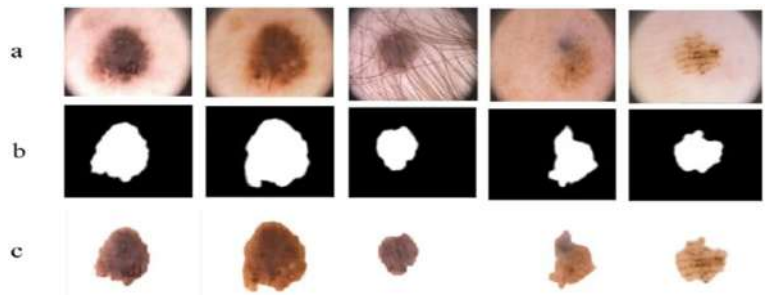

Figure 7: Sample images from the dataset of Topic G: (a) skin lesions (b) segmentation masks (c) segmented skin lesions.

Problems G1 and G2

For the given datasets G1 and G2, it is intended to develop a lesion segmentation algorithm that relies solely on classical computer vision techniques:

1. Analyse the images provided in the dataset and apply appropriate preprocessing techniques to address dataset-specific challenges (e.g. colour spae transformations, illumination normalization, etc.).
2. Implement and compare different classical segmentation approaches, and study the influence of the tuning parameters in the result.
3. Apply post-processing techniques (e.g. morphological operations) to refine segmentation masks.
4. Compare your automated segmentation masks against expert annotations using standard metrics (justify your choice of performance metrics).
5. Analyse failure cases and discuss which image characteristics most challenge your methods.

Topic H: Historical architectural surface crack detection

Historical building preservation requires continuous monitoring of structural integrity. This project (see reference) uses the Historical Building Crack 2019 (HBC2019) dataset, which contains real-world images from the Mosque of Amir al-Maridani in Cairo, Egypt, a historical building from 1339-40 CE suffering from cracking problems. The dataset comprises 3,886 RGB images (256 × 256 pixels): 757 crack images and 3,139 non-crack images, captured using a Canon EOS camera over 2018-2019. This dataset presents realistic challenges found in historical building inspections, including dust, variable illumination, blur, fine textures, wood grain patterns, crack-like features (e.g., separators), and other visual noise that make detection difficult. This work intends to constitute a contribution to the problem solution based on a set of provided images, exploiting studied techniques of image processing. A sample image of the dataset to be provided is shown in Figure 8.

Figure 8: Sample images from the dataset of Topic H.

Problems H1 and H2

For the given datasets H1 and H2, it is intended to develop a binary classifier that distinguishes cracked from noncracked surfaces using only classical computer vision techniques:

1. Analyse the images provided in the dataset and apply appropriate preprocessing techniques to address dataset-specific challenges (e.g. colour space transformations, illumination normalization, etc.).
2. Implement and compare different classical computer vision techniques to extract discriminative features, and study the influence of the tuning parameters in the result.
3. Design a rule-based or threshold-based classifier (cracked or not cracked) using your extracted features.
4. Compare your automated binary classifier against expert annotations using standard metrics (justify your choice of performance metrics).
5. Analyse failure cases and discuss which image characteristics most challenge your methods.